Topic: Multiplying Algebraic Expressions

EQ: How do I simplify algebraic expressions with multiplication?

| Question | Working Out | Explanation |
| --- | --- | --- |
| $4 \times 3a$ | ① $4 \times 3 = 12$<br>② $a = a$<br>③ $\boxed{12a}$ | ☆① multiply the **numbers** first<br>② multiply any variables (this only had (a) so nothing changed)<br>③ combine the numbe[rs] and the variables |
| $3d \times 4e$ | ① $3 \times 4 = 12$<br>② $d \times e = de$<br>③ $\boxed{12de}$ | ☆① same↑<br>② (d) and (e) are not like variables so I can only write them as $d \times e$ or $de$ ← simplified w/o × sym[bol]<br>③ combined the number and both variables without the × symbo[l] |
| $6a \times 3a$ | ① $6 \times 3 = 18$<br>② $a \times a = a^2$<br>③ $\boxed{18a^2}$ | ☆① same↑<br>② (a) and (a) are like terms so can be simplified using exponents<br>③ combined |
| $5mn \times -2m^2n$ | ① $5 \times -2 = -10$<br>② $m \times m^2 = m^3$<br>③ $n \times n = n^2$<br>④ $-10m^3n^2$<br>These steps were split up b/c not like terms | ☆① same↑ (careful with negatives)<br>② Index law 1<br>$a^m \times a^n = a^{m+n}$<br>(m) by itself is the same as $m^1$<br>$m^1 \times m^2 = m^{2+1} = m^3$<br>③ $n \times n = n^2$<br>④ combined all parts |

Summary:

To simplify algebraic equations with multiplication we have to ____________________.

If there are two variables the same we have to use __________.